

HUGHENDEN, Australia

WMO: 943430

Lat: 20.819S

Lon: 144.233E

Elev: 1041

StdP: 14.15

Time Zone: 10.00 (E10)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 40.8	(c) 44.1	(d) 12.7	(e) 10.9	(f) 63.1	(g) 15.3	(h) 12.4	(i) 63.6	(j) 20.1	(k) 67.5	(l) 18.5	(m) 68.3	(n) 2.9	(o) 150	(p) 0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 12	(b) 22.4	(c) 102.9	(d) 66.7	(e) 100.7	(f) 66.6	(g) 98.8	(h) 66.5	(i) 75.5	(j) 86.1	(k) 74.6	(l) 85.0	(m) 73.7	(n) 84.1	(o) 10.1	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	126.9	79.4	71.9	122.6	79.3	70.8	118.0	79.3	39.9	86.0	39.0	84.7	38.1	83.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.0	(b) 18.4	(c) 16.8	(e) 35.4	(f) 105.8	(g) 2.3	(h) 2.7	(i) 33.7	(j) 107.8	(k) 32.4	(l) 109.4	(m) 31.1	(n) 110.9	(o) 29.4	(p) 112.9		
			DB													
			WB													
				29.6	77.7	3.0	1.8	27.4	79.0	25.6	80.1	24.0	81.1	21.8	82.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.4	84.6	83.3	81.9	77.3	70.5	64.6	64.2	66.7	74.4	80.3	83.8	86.1
	DBStd	9.27	3.96	3.92	3.41	3.98	5.73	6.10	6.23	6.20	5.48	5.25	4.05	3.96
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	258	0	0	0	1	21	83	93	55	4	1	0	0
	CDD50	9649	1073	932	989	820	635	438	440	518	733	939	1014	1118
	CDD65	4432	608	512	524	371	191	71	68	108	288	474	564	653
	CDH74	54726	7779	6296	6134	3790	1653	564	645	1361	3519	6303	7526	9156
	CDH80	28286	4214	3309	3060	1626	471	84	106	411	1664	3526	4356	5459
(14) Wind	WSAvg	9.1	9.4	9.4	9.5	9.4	8.8	8.1	8.1	8.3	9.2	9.9	9.9	9.6
(15) Precipitation	PrecAvg	17.7	4.1	3.8	2.1	0.9	0.7	0.6	0.4	0.4	0.2	0.9	1.5	2.7
	PrecMax	36.3	24.8	11.1	9.9	9.3	4.4	4.2	4.4	4.7	3.0	7.6	6.7	8.9
	PrecMin	9.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
	PrecStd	7.0	4.4	2.9	2.4	1.8	1.1	1.1	0.8	0.9	0.7	1.5	1.7	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.5	103.1	99.5	94.6	91.4	85.7	86.5	91.7	98.2	103.1	103.7	105.6
		MCWB	68.7	68.3	67.8	67.1	64.9	63.3	62.7	61.8	61.8	64.5	66.1	67.0
	2%	DB	100.9	99.0	96.7	92.3	87.5	82.3	82.8	87.9	94.9	99.7	101.0	103.2
		MCWB	68.6	68.8	67.9	65.8	63.3	62.2	60.8	60.5	61.5	63.7	65.5	66.7
	5%	DB	98.0	96.5	94.5	90.3	85.0	79.9	80.3	84.5	91.8	97.2	98.8	100.7
		MCWB	68.8	69.2	67.9	64.9	62.4	60.9	59.3	59.1	60.9	63.4	65.2	66.9
	10%	DB	95.4	94.4	92.4	88.3	82.4	77.3	77.8	81.5	89.0	94.6	96.5	98.4
		MCWB	69.1	69.1	68.0	64.7	61.8	59.5	58.1	57.6	60.5	63.1	65.0	67.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.7	76.9	75.3	73.5	68.3	67.1	66.3	66.4	69.5	69.9	74.2	75.2
		MCDB	87.9	88.0	85.1	82.7	83.0	78.7	76.9	82.9	81.3	85.4	85.8	86.1
	2%	WB	75.6	75.6	73.7	71.2	66.8	64.8	64.1	63.7	67.4	68.0	71.9	73.8
		MCDB	86.3	86.0	84.1	82.8	80.3	76.7	77.8	79.4	80.9	87.4	83.3	85.5
	5%	WB	74.6	74.5	72.6	69.8	65.5	63.3	62.2	61.7	65.2	66.8	70.5	72.8
		MCDB	85.0	84.7	83.8	81.6	78.8	75.5	75.3	78.3	81.4	86.8	84.2	85.3
	10%	WB	73.8	73.5	71.7	68.3	64.0	61.9	60.5	60.0	63.0	65.8	69.3	71.9
		MCDB	83.9	83.7	83.7	81.0	77.8	73.8	73.8	76.2	82.4	86.1	84.9	85.3
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	19.6	20.3	21.5	23.1	24.9	26.6	28.5	27.8	26.5	23.7	22.4
		MCDBR	24.1	24.0	23.7	24.1	24.6	25.4	26.9	30.1	30.7	29.3	26.9	26.4
		MCWBR	6.3	6.3	6.6	7.8	9.2	10.2	11.1	11.9	10.7	9.6	7.9	7.8
	5% WB	MCDBR	16.9	17.0	17.5	17.6	18.7	20.5	21.2	23.5	22.3	23.5	20.7	19.6
		MCWBR	4.8	5.3	5.2	5.9	7.2	8.0	8.6	8.7	7.7	7.2	6.4	5.8
(40) Clear-Sky Solar Irradiance	taub		0.375	0.368	0.352	0.337	0.320	0.308	0.300	0.316	0.350	0.375	0.383	0.392
	taud		2.534	2.552	2.577	2.580	2.562	2.587	2.592	2.520	2.434	2.382	2.409	2.428
	Ebn at Noon		307	305	303	295	288	286	292	298	299	300	303	302
	Edh at Noon		36	34	32	30	28	26	27	31	36	40	40	39
(44) All-Sky Solar Radiation	RadAvg		2194	2176	2025	1820	1525	1368	1518	1814	2132	2325	2377	2322
	RadStd		178	152	172	158	93	107	71	87	126	144	154	169

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page